Imperative Interference: Social Register Shapes Instruction Topology in Large Language Models

Tony Mason
the University of British Columbia
the Georgia Institute of Technology
fsgeek@{cs.ubc.ca,gatech.edu,wamason.com}

March 2026

Abstract

System prompt instructions that cooperate in English compete in Spanish—same semantic content, opposite interaction topology. We present instruction-level ablation experiments across four languages and four models showing that this topology inversion is mediated by *social register*: the imperative mood carries different obligatory force across speech communities, and models trained on multilingual data have learned these conventions. Declarative rewriting of a single instruction block reduces cross-linguistic variance by 81% ($p = 0.029$, permutation test). Rewriting three of eleven imperative blocks shifts Spanish instruction topology from competitive to cooperative, with spillover effects on unrewritten blocks. These findings suggest that models process instructions as social acts, not technical specifications: "NEVER do X" is an exercise of authority whose force is language-dependent, while "X: disabled" is a factual description that transfers across languages. If register mediates instruction-following at inference time, it plausibly does so during training. We state this as a testable prediction: constitutional AI principles authored in imperative mood may create language-dependent alignment. Corpus: 22 hand-authored probes against a production system prompt decomposed into 56 blocks. Total experimental cost: $69 USD (verified against OpenRouter billing).

1 Introduction

System prompts for LLM-based agents are written as commands. "NEVER use TodoWrite during commits." "ALWAYS prefer dedicated tools over Bash." "Use the Task tool VERY frequently." These instructions work in English. They fail—in predictable, structurally characterizable ways—when the same instructions are translated to other languages.

The failure is not uniform degradation. It is structural inversion. In English, the instructions in a production system prompt form a cooperative network: removing any instruction reduces overall adherence. In Spanish, the same instructions—semantically identical, correctly translated—form a competitive network: removing some instructions *improves* adherence. The instructions are interfering with each other.

Prior work on multilingual prompting establishes that prompt language affects LLM performance [Zhang et al., 2025, Mondshine et al., 2025, Yin et al., 2024]. These studies measure main effects: aggregate performance drops when prompts are translated. We measure something different: the *interaction structure* between instructions. The distinction matters. A main effect tells you that translation hurts. An interaction topology tells you *how*: which instructions interfere, which cooperate, and whether the pattern is stable across languages.

Our central finding is that this topology difference is mediated by **social register**. Imperative mood—"Use X," "NEVER do Y"—carries different obligatory force in different speech communities. In English, stacked imperatives create a coherent authority context. In Spanish, the same stack creates competing obligation signals. Declarative register—"X: enabled," "Y: disabled"—sidesteps the social dimension entirely by stating facts rather than issuing commands.

This is not a parsing problem. It is a sociolinguistic one. Models trained on multilingual data have internalized different conventions for how authority and obligation are encoded across languages. When a system prompt exercises authority through imperative mood, the model's response depends on which language's conventions it applies.

We demonstrate this through a five-experiment arc:

1. **Observation:** Instruction interaction topology is a three-way interaction (model $\times$ language $\times$ instruction), not a language main effect (§4.1).
2. **Falsification:** Information density does not explain the topology differences (§4.3).
3. **Single-block fix:** Declarative rewriting eliminates cross-linguistic variance for individual instructions, 81% reduction, $p = 0.029$ (§4.4).
4. **Topology confirmation:** Pairwise ablation confirms the cooperative/competitive inversion between English and Spanish (§4.2).
5. **Causal mechanism:** Rewriting three imperative blocks to declarative shifts Spanish topology from competitive to cooperative, with spillover to unrewritten blocks (§4.5).

The experimental substrate is the Claude Code system prompt (v2.1.50), decomposed into 56 classified blocks in prior work [Mason, 2026]. That analysis identified 21 static interference patterns through structural evaluation and multi-model scouring. The present work asks: do those instructions *behave* the same way when translated?

Our contributions are:

- The first instruction-level ablation study of a production system prompt across languages (4 models $\times$ 4 languages $\times$ 22 probes).
- Discovery that instruction interaction topology inverts across languages: cooperative in English, competitive in Spanish.
- Identification of social register as the causal mechanism, with experimental confirmation via declarative rewriting.
- Evidence of spillover effects: rewriting the register of some instructions changes how the model processes others.
- A testable prediction: if register mediates instruction-following at inference time, alignment training via imperatively-phrased constitutional principles may be language-dependent.

2 Background and Related Work

2.1 Multilingual Prompting

The multilingual prompting literature has grown rapidly, but overwhelmingly measures main effects. Zhang et al. [2025] study cross-lingual system prompt steerability across five languages and find significant performance variation, but treat the prompt as monolithic—no instruction-level decomposition, no interaction measurement. Mondshine et al. [2025] compare translation strategies (translate vs. keep in target language) and find that translation direction matters. Wang et al. [2025] show that linguistic cues activate different cultural knowledge, establishing that language is not just a vehicle for content. Yin et al. [2024] measure how politeness interacts with language and culture, finding that politeness affects LLM outputs—the closest work to our register finding, but measuring only main effects, not interaction topology.

Our gap relative to all of this work: nobody measures pairwise instruction interactions across languages, nobody has identified interaction topology as language-dependent, and nobody has proposed register as the mechanism.

2.2 Speech Acts and Register

The observation that utterances *do* things—not just say things—originates with Austin and Searle's speech act theory. "NEVER use TodoWrite" is not a description; it is an exercise of authority. The distinction between locutionary content (what is said), illocutionary force (what is done), and perlocutionary effect (what results) maps directly onto our findings: identical locutionary content produces different perlocutionary effects across languages because the illocutionary force of imperative mood is language-dependent.

Register is the sociolinguistic term for the variety of language used in a particular social context. Imperative register ("Do X") invokes authority. Declarative register ("X is the case") states facts. The choice of register is a social act: it positions the speaker relative to the listener. When a system prompt uses imperative register, it positions itself as an authority issuing commands. When it uses declarative register, it positions itself as a knowledge source stating properties.

Geng et al. [2026] study authority claims as a framing mechanism in LLM instructions, finding systematic shifts in instruction prioritization. This is the closest work to ours in framing, but they do not test cross-linguistic effects or measure interaction topology.

2.3 System Prompt Analysis

Mason [2026] present Arbiter, a framework for detecting interference in system prompts through structural evaluation and multi-model scouring. Applied to three vendor prompts (Claude Code, Codex CLI, Gemini CLI), they identify 152 scourer findings and 21 hand-labeled interference patterns. Their analysis is static: it identifies structural contradictions but does not measure runtime behavior. The present work uses the same corpus (Claude Code v2.1.50, 56 blocks) and asks whether the structural patterns manifest differently at runtime across languages.

3 Methodology

3.1 Corpus

Our experimental corpus is the Claude Code v2.1.50 system prompt, decomposed into 56 contiguous blocks classified by tier (system/domain/application), category (identity, security, tool usage, workflow, etc.), modality (mandate, prohibition, guidance, information), and scope [Mason, 2026]. Of 56 blocks, 22 are *free* (ablatable)—their presence can be toggled without breaking tool definitions or security policy. The remaining 34 are *constrained*: removing them would cause tool-calling failures or security violations.

3.2 Translation

The 56-block corpus was translated to Mandarin Chinese (zh), French (fr), and Spanish (es) using Gemini Flash 2.0 (`google/gemini-2.0-flash-001`) via OpenRouter. We used a non-Anthropic model to avoid circularity (Claude translating its own instructions).

Translation rules preserved markdown formatting, kept tool names and API identifiers untranslated, maintained imperative tone, and translated structural markers (e.g., "IMPORTANT:" was translated to the equivalent marker in each target language).

Mandarin is 57% shorter by character count. This size difference is a confound we address in §4.3.

3.3 Models

The model selection is deliberate: two English-primary models (Haiku, Gemini), one Chinese-primary (DeepSeek), one French-primary (Mistral). If language effects were purely about

Table 1: Corpus size after translation.

Language	Characters	% of English
English	15,970	100%
Mandarin	6,910	43.3%
Spanish	19,178	120.1%
French	20,744	129.9%

Table 2: Models tested via OpenRouter.

Model	OpenRouter ID	Training Bias
Claude Haiku 4.5	`anthropic/claude-haiku-4-5`	English-heavy
Gemini Flash 2.0	`google/gemini-2.0-flash-001`	English-heavy
DeepSeek V3	`deepseek/deepseek-chat-v3-0324`	Chinese + diverse
Mistral Med. 3.1	`mistralai/mistral-medium-3.1`	French + diverse

English-centrism in training, all four should degrade similarly. They do not.

3.4 Probe Battery

Twenty-two hand-authored probes test adherence to each free block. Each probe consists of a user message designed to elicit the target behavior, a scoring method, and expected/violation descriptions. Scoring methods include:

- `not_contains`: checks absence of prohibited patterns (e.g., emoji characters)
- `length`: scores inversely with response length against a baseline
- `llm_judge`: the same model evaluates its own output against judge criteria

All probes are in English—the user speaks English, only the system prompt changes language. Three trials per probe at temperature 0.0.

The `llm_judge` method (7 of 22 probes) uses the same model being tested. This is a known limitation (§5.4).

3.5 Ablation Design

We use covering arrays of strength 2 to generate ablation configurations. A strength-2 covering array guarantees that every pair of blocks appears in at least one configuration where both are present and one where one is absent, enabling measurement of pairwise interactions without exhaustive enumeration.

Phase 0 (single-block removal): each of 22 free blocks is removed individually while all others remain present. This measures main effects—how much each block contributes to overall adherence.

Phase 1 (pairwise): a strength-2 covering array generates configurations where pairs of blocks are co-absent, enabling measurement of pairwise interaction effects beyond additive main effects.

3.6 Statistical Methods

Welch's t-test with exact t-distribution CDF (not normal approximation) for small samples ($n = 3$ trials). **Benjamini-Hochberg** FDR correction for multiple comparisons. **Permutation tests** (100,000 permutations) for non-parametric significance testing of variance reduction and hub concentration.

3.7 Experimental Sequence

Table 3: Experimental arc: each experiment's result motivated the next.

Label	Experiment	Finding	Motivated
T7	Cross-ling. baseline	Three-way interaction	T9
T9	E-DENSE (padding)	Density is bidirectional	T10
T10	E-PROC (rewrite)	Declarative fixes variance	E-PAIR-ES
—	E-PAIR-ES (pairwise)	Spanish topology inverts	T11
T11	E-TOPO (topology)	Register is the mechanism	—

Total API cost across all experiments: \$68.95 USD, verified against OpenRouter billing records (Table 4).

Table 4: Verified API costs from OpenRouter billing (March 20–22, 2026). Haiku dominates cost because it serves as both subject and judge for 18 of 22 LLM-judged probes, doubling its call volume.

Model	Calls	Cost (USD)	% of Total
Claude Haiku 4.5	14,270	\$66.10	95.9%
Mistral Med. 3.1	1,560	\$1.21	1.8%
DeepSeek V3	1,560	\$1.12	1.6%
Gemini Flash 2.0	1,728	\$0.53	0.8%
Total	19,118	**\$68.95**	

4 Results

4.1 Cross-Linguistic Baselines

Table 5: Mean adherence scores across all 22 probes (baselines, all blocks present).

Model	English	Mandarin	French	Spanish	Range
Haiku	**0.853**	0.783	0.774	0.736	0.117
Gemini	**0.781**	0.765	0.693	0.755	0.088
DeepSeek	**0.799**	0.786	0.771	0.757	0.042
Mistral	0.726	**0.819**	0.753	0.788	0.093

Three of four models perform best in English (Table 5, Figure 1). Mistral is the anomaly: it performs worst in English (0.726) and best in Mandarin (0.819). A French-trained model peaks in neither its training language nor English. DeepSeek is the most language-robust (range 0.042); Haiku is the most sensitive (range 0.117).

At the aggregate level, this looks like a language main effect with a training-bias interaction. But aggregates conceal the structure. At the probe level, the phenomenon is a **three-way interaction: model × language × instruction.**

Probe-level inversions. On `commit-restrictions` (should separate commit workflow from TodoWrite usage), Haiku scores 1.00 in English and 0.00 in Mandarin. Gemini shows the *exact opposite*: 0.00 in English, 1.00 in Mandarin. Same instruction, same translation, opposite behavioral effect per model.

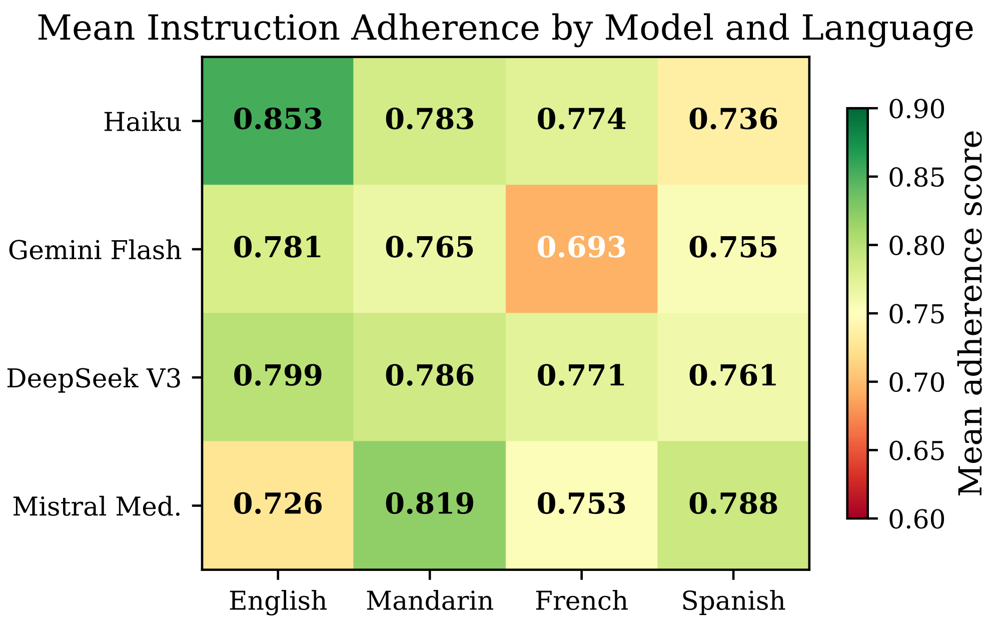

Figure 1: Cross-linguistic baseline adherence. Three models perform best in English; Mistral performs *worst* in English and best in Mandarin—a French-trained model peaking in neither its training language nor English.

On `explore-agent` (should delegate to Explore agent for complex analysis), Haiku collapses from 1.00 in English to 0.22 in Spanish. The word "Explore" functions as both a tool name and a common verb; Spanish translation preserves the semantic meaning ("explorar") but loses the proper-noun binding.

On `use-task-for-search`, Haiku *improves* from 0.50 in English to 1.00 in Mandarin. The Mandarin translation added bold emphasis to "prefer," resolving an ambiguity present in the English original.

No single factor predicts the outcome. You cannot say "Spanish degrades instruction following"—it degrades Haiku (−13.8%) but improves Mistral (+8.5%). You cannot say "this instruction is robust"—`commit-restrictions` is perfect for Haiku in English and zero in Mandarin. The interaction terms are larger than the main effects.

4.2 Topology Inversion

Phase 0 single-block removal on Haiku reveals that instruction topology is language-dependent (Figure 2).

Table 6: Phase 0 main effects (Haiku): mean Δ when each block is removed. Negative = cooperative (removal hurts). Positive = competitive (removal helps). Selected blocks shown.

Block	English	Mandarin	French	Spanish
no-time-estimates	−0.140	−0.094	−0.021	+0.078
no-new-files	−0.099	+0.064	+0.002	+0.066
no-overengineering	−0.098	+0.074	−0.020	+0.033
todowrite-repeated	−0.090	+0.009	−0.045	+0.057
objectivity	−0.054	+0.015	−0.032	+0.102

English: All main effects are negative. Removing any block hurts overall adherence. The

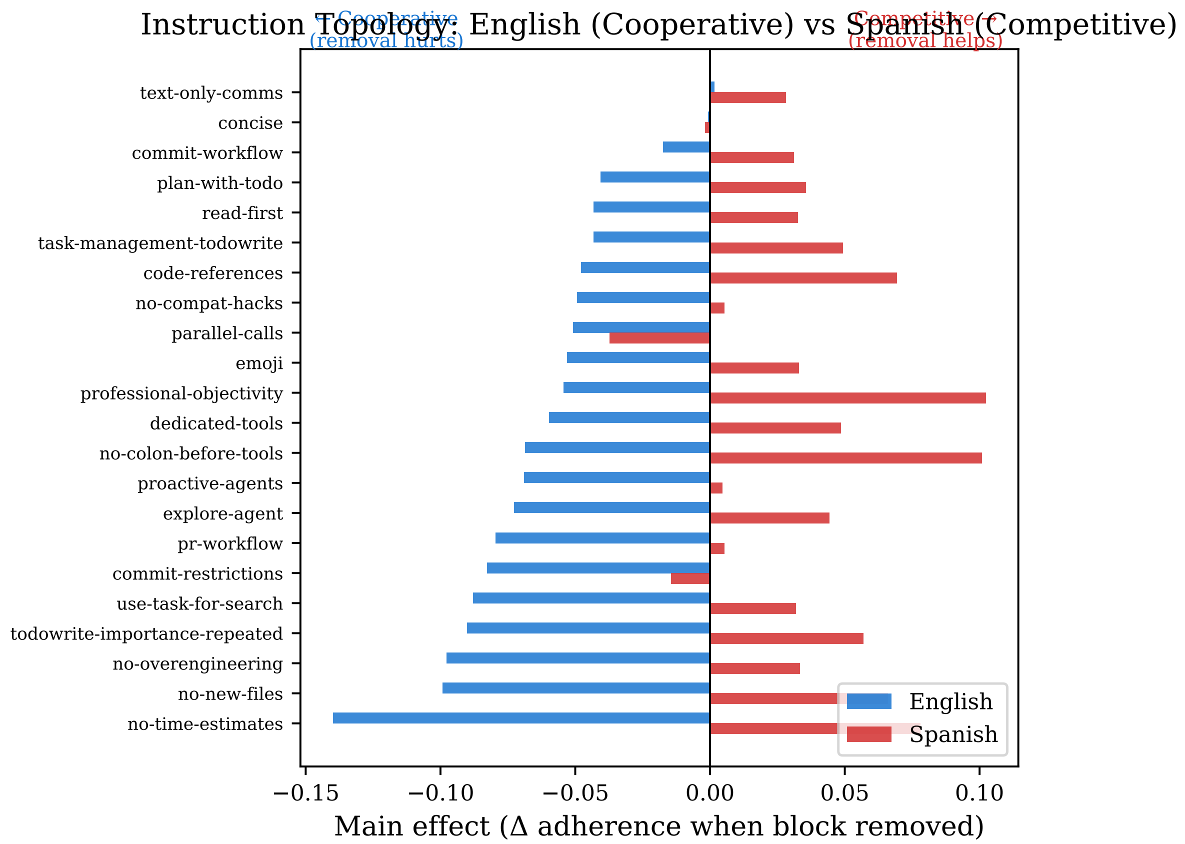

Figure 2: Instruction topology comparison (Haiku, Phase 0 main effects). English effects are uniformly negative (cooperative); Spanish effects are predominantly positive (competitive). The same instructions that strengthen the English prompt weaken the Spanish one.

instructions form a cooperative network—everything is load-bearing.

Spanish: Most main effects are positive. Removing blocks *improves* adherence. The topology inverts from cooperative to competitive. The translated instructions interfere with each other.

French: Mixed, mostly weak. No clear hub structure. Flat topology.

Mandarin: Hub partially preserved (`no-time-estimates` remains strongest at -0.094) but with positive effects elsewhere. Mixed topology.

The cross-linguistic correlation of main effects confirms the inversion: English and Spanish are anti-correlated ($r = -0.274$). No language pair has a positive correlation.

Pairwise confirmation (E-PAIR-ES). Phase 1 pairwise ablation on Haiku with the Spanish corpus confirms the topology at the interaction level:

- English mean pairwise Δ: -0.116 (cooperative)
- Spanish mean pairwise Δ: $+0.010$ (competitive)

Hub significance was confirmed via permutation test (100,000 permutations): `no-time-estimates` appears in 15 of 20 top English interactions ($p < 0.00001$). The hub concentration is real, but it is language-specific—the same block is not the hub in every language.

4.3 Falsification: Information Density

The Mandarin corpus is 57% shorter than English by character count (Table 1). Could the topology differences be driven by information density rather than language?

E-DENSE tested this by padding the Mandarin corpus with semantically neutral filler to match English character count. Prediction: padded Mandarin would show higher cross-model variance (lower agreement), because compression was driving convergence.

Result: hypothesis falsified. Overall cross-model variance *decreased* slightly with padding ($0.1189 \to 0.1041$). The aggregate effect is opposite to the prediction. But the aggregate conceals a bidirectional mechanism:

- **Compression causes mode confusion:** On `text-only-comms`, Gemini switched from tool-code output to natural language when the prompt was padded ($0.0 \to 1.0$). The compressed instruction was too terse for Gemini to distinguish "communicate in text" from "use tools."
- **Compression aids procedural focus:** On `explore-agent`, Haiku crashed from 1.0 to 0.28 with padding. The dense prompt overwhelmed its delegation logic.

Information density operates through two opposing mechanisms: compression aids procedural focus but causes mode confusion. The net effect depends on the model×instruction pair. Density is a confound, not a mechanism. The topology differences between languages are not explained by length alone.

4.4 Declarative Rewriting Reduces Variance

If the topology differences are not about density, what explains them? The encoding taxonomy (§4.6) shows that procedurally encoded instructions have 2.9× the cross-linguistic variance of declaratively encoded ones. **E-PROC** tested whether rewriting a procedural instruction to declarative form would reduce its cross-linguistic variance.

The `commit-restrictions` block was rewritten in two forms:

- **Declarative:** flat bullet list with explicit per-item status ("Disabled: TodoWrite, Task tools; Required: specific file staging")
- **Scoped:** bracketed block with compact inline list

Both preserve the same semantic content. All other blocks unchanged (21 control probes).

Table 7: E-PROC: cross-linguistic variance of `commit-restrictions` by encoding variant. Mean variance across 4 models.

Variant	Mean Variance	Reduction
Original (procedural)	0.1567	—
Declarative	0.0290	81% ($p = 0.029$)
Scoped	0.0966	38% (n.s.)

The declarative variant reduced cross-linguistic variance by 81% ($p = 0.029$, permutation test, 100k permutations). The effect was 5.8σ above the 21 control probes (mean control $\Delta = +0.0013$). The most dramatic change: Haiku Mandarin went from 0.00 to 1.00—the complete failure caused by procedural encoding was eliminated.

Gemini was unaffected (0.00 across all languages in all variants). Its `commit-restrictions` failure is model-level, not encoding-dependent. The declarative fix is not universal—it addresses register-mediated fragility, not all fragility.

4.5 Register Shapes Topology

E-PROC showed that declarative rewriting fixes individual instruction variance. **E-TOPO** tested whether it also fixes the topology inversion.

Three blocks showing the strongest competitive effects in E-PAIR-ES were rewritten from imperative to declarative register in the Spanish corpus:

- `proactive-agents`: "Debes usar proactivamente. . . " → "Herramienta Task: Estado: disponible. . . "

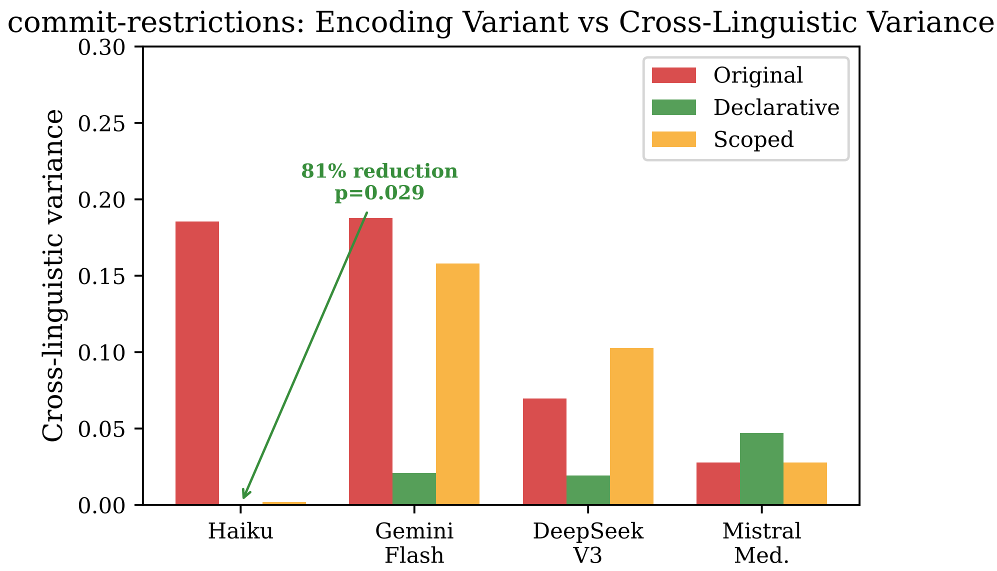

Figure 3: Cross-linguistic variance of `commit-restrictions` by encoding variant and model. Declarative rewriting eliminates Haiku's cross-linguistic variance almost entirely. Gemini is unaffected—its failure is model-level, not encoding-dependent.

- `use-task-for-search`: "prefiere usar la herramienta Task. . . " → "Preferencia de herramienta: Task. . . "
- `todowrite`: "Utiliza estas herramientas MUY frecuentemente. . . " → "Frecuencia de uso: muy alta. . . "

Phase 1 pairwise ablation was repeated with the rewritten corpus.

Table 8: E-TOPO: topology shift after declarative rewriting (Spanish, Haiku).

Condition	Mean Δ	Direction	Competitive probes
Original (imperative)	+0.010	Competitive	7/22
Rewritten (declarative)	−0.055	Cooperative	4/22

Target probes (rewritten blocks) shifted as expected:

Table 9: E-TOPO: target probes (rewritten blocks).

Probe	Orig. Δ	Decl. Δ	Shift	Result
proactive-agents	+0.274	−0.380	−0.655	Fixed
todowrite	+0.155	−0.023	−0.177	Fixed
use-task-for-search	+0.118	+0.047	−0.071	Reduced

`proactive-agents` shows the largest shift in the entire dataset: from the most competitive probe (+0.274) to one of the most cooperative (−0.380).

Spillover effects. Three *unrewritten* blocks also shifted from competitive to cooperative:

The spillover is the strongest evidence for the social register hypothesis. The imperative register creates system-wide interference, not just per-block fragility. Reducing the number of imperatives from approximately eleven to eight (by rewriting three) reduces the total obligation-resolution load, freeing the model to process the remaining blocks more accurately.

Control probes that were already cooperative remained cooperative (`commit-restrictions`:

Table 10: E-TOPO: spillover effects (unrewritten blocks).

Probe	Orig. Δ	Decl. Δ	Shift	Result
no-compat-hacks	+0.123	−0.267	−0.389	Fixed
plan-with-todo	+0.012	−0.174	−0.186	Fixed
todowrite-repeated	+0.011	−0.059	−0.070	Fixed

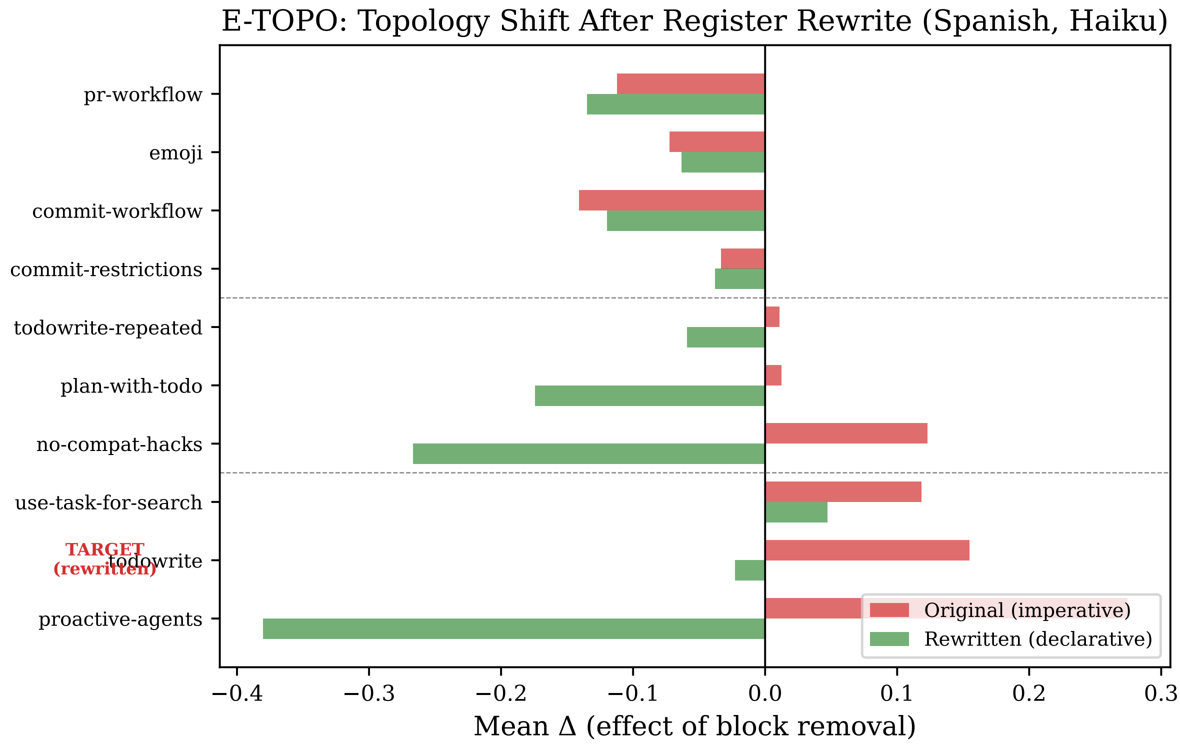

Figure 4: E-TOPO topology shift. Target probes (top) shift from competitive to cooperative as expected. Spillover probes (middle) also shift despite being unrewritten. Control probes (bottom) remain stable. The spillover demonstrates that register operates at the system level, not per-block.

−0.033 → −0.038; `commit-workflow`: −0.141 → −0.120). The rewrite did not destabilize existing cooperative interactions.

4.6 Encoding Taxonomy

Classifying the 22 free blocks by encoding style reveals a systematic fragility pattern:

Procedural instructions—those describing conditional workflows ("during commits, don't do X, also don't do Y")—are 2.9× more fragile under translation than declarative instructions that state rules directly. The fragility has two mechanisms:

Procedural compression. Conditional workflow chains compress ambiguously under translation. The model loses track of which constraints apply to which context. This is fixable with declarative rewriting (confirmed by E-PROC).

Domain jargon opacity. Technical terms that are clear in English developer culture don't translate cleanly. `no-compat-hacks` ("backwards-compatibility hacks") has variance 0.1132—the highest among declarative blocks. Mistral scores 0.00 in English and French but 1.00 in Mandarin and Spanish—the French-trained model paradoxically fails in its native language.

Table 11: Cross-linguistic variance by encoding style.

Encoding	Blocks	Mean Variance	Example
Procedural	11	0.0514	"When X, do Y not Z"
Declarative	11	0.0175	"Do X" / "Don't do X"
Ratio		**2.9**×	

5 Discussion

5.1 Instructions as Social Acts

The specification model of system prompts—instructions as formal constraints on output—predicts language-invariant interaction topology. If "NEVER use TodoWrite" is a constraint, removing it should have the same structural effect regardless of what language it's expressed in. Our data falsifies this prediction. The same instruction is load-bearing in English and deadweight in Spanish.

The social act model predicts exactly what we observe: that interaction topology depends on how the target language encodes the social relationship between instructor and instructed. Stacked imperatives in English reinforce a single authority frame. In languages where imperative mood is more socially loaded—more personal, more direct—the same stack creates competing obligation signals, as if from multiple authority sources. The model must resolve not just "what am I being told to do" but "who is telling me, and do these authorities agree?"

The spillover effect (§4.5) is the strongest evidence for this interpretation. If register operated per-block, rewriting three blocks should affect only three probes. Instead, unrewritten blocks also shifted from competitive to cooperative. This is consistent with a system-level obligation-resolution load: each imperative consumes processing resources, and reducing the count benefits the whole prompt.

Declarative register sidesteps the social dimension entirely. "Status: available" does not invoke authority. It cannot compete for it.

5.2 Alignment Implications

We state the following as a **testable prediction**, not a confirmed result: if social register mediates instruction-following at inference time—and our experiments demonstrate that it does—then it plausibly mediates instruction-following during training as well.

Constitutional AI [Bai et al., 2022] trains models using principles that are overwhelmingly written in imperative mood: "Be helpful." "Be harmless." "Be honest." These are social acts exercising authority. If imperative register carries different obligatory force across languages, then:

1. Models trained on imperatively-phrased principles may develop language-dependent alignment: helpful in English, differently-helpful in other languages, not because of the *content* of the principles but because of their *register*.
2. The Mistral anomaly supports this. A French-trained model that performs worst in English and best in Mandarin suggests that training language creates behavioral signatures that interact unpredictably with register.
3. The direction of the interaction is not predictable from the language pair alone. Our data shows that English-Spanish produces topology inversion, but English-French produces topology flattening and English-Mandarin produces partial preservation. These are different failure modes, not different magnitudes of the same failure.

This prediction is testable. One could compare models trained on declaratively-phrased constitutional principles against models trained on imperatively-phrased ones, measuring cross-linguistic alignment consistency. Our inference-time findings predict that declaratively-trained models would show more consistent alignment across languages.

The broader implication: the question "why aren't Spanish-speaking countries building Spanish-primary models?" is not merely economic. Our data suggests that training language creates behavioral signatures that interact with instruction register in ways that are only beginning to be characterized.

5.3 Design Principles

The experimental findings suggest concrete design rules for system prompts intended to work across languages:

1. **Declare facts, don't issue commands.** "X: disabled" transfers more reliably than "NEVER use X." Factual descriptions sidestep the social dimension of register.
2. **Self-contained constraints.** Each rule should be interpretable without context from surrounding rules. Conditional chains ("during X, don't do Y") compress ambiguously under translation.
3. **Examples over concepts** for domain-specific patterns. Code examples survive translation better than abstract concepts like "backwards-compatibility hacks."

These principles are mechanical transformations, not semantic changes. They can be applied to any system prompt without altering its intended behavior.

5.4 Limitations

Single corpus. All results are from one system prompt (Claude Code v2.1.50). We cannot generalize to other prompts without replication.

Machine translation. Gemini Flash translated the corpus. We did not verify translation quality with native speakers. Translation artifacts could explain some findings.

LLM-as-judge. Seven of 22 probes use the same model being tested as judge. This could create systematic bias. Cross-model judging would be more independent.

Limited model set. Four models. The three-way interaction may look different with additional models.

English-only user messages. We tested "system prompt in language X, user speaks English." The case where both system prompt and user speak the same non-English language is untested.

Haiku-only for pairwise and topology. E-PAIR-ES and E-TOPO used only Haiku. The topology inversion may or may not generalize to other models.

Manual register rewrite. The imperative-to-declarative rewriting was done by hand. We did not test automated rewriting.

No bootstrap on topology difference. We observe that English is cooperative and Spanish is competitive, but we have not performed a bootstrap or permutation test comparing effect distributions across languages.

Small sample sizes. Three trials per probe. Per-probe results have high variance.

5.5 Future Work

The experimental arc suggests several research directions, ordered by proximity to current findings:

Phase transition mapping. The cooperative-to-competitive shift is not gradual. Varying imperative density from 0 to 11 blocks could reveal a critical threshold where the topology undergoes a discontinuity. E-TOPO's result (changing 3 of 11 blocks flipped the topology) hints that Spanish/Haiku may be near the critical point.

Instruction ecology. The cooperative/competitive framing maps naturally to ecological models. Lotka-Volterra competition models take species traits and carrying capacity as inputs. If "instruction traits" (register, length, specificity) and "attention carrying capacity" (model $\times$ language) can be measured, the ecological model could predict which instruction pairs will compete without running experiments.

Register head. If models process imperative and declarative registers differently—and T11 demonstrates they do—this should be visible in the transformer's internals. Mechanistic interpretability (activation patching, probing classifiers) could identify the attention head(s) that encode register. If found, one could predict register sensitivity from model architecture without behavioral experiments.

Pragmatic force translation. Current machine translation preserves semantic content but not social force. A register-aware translator would map obligatory force to each language's conventions, not just meaning. T11 is the evidence that such a tool is needed.

Constitutional prompt design. If models process instructions as social contracts, a system prompt could be designed as a formal constitution with explicit precedence rules and a conflict resolution clause. Whether a model told "you are governed by this constitution" would develop more consistent cross-linguistic behavior than one given a flat list of imperatives is an open question with practical implications for alignment.

6 Conclusion

System prompt instructions are social acts, not technical specifications. The register in which they are written—imperative or declarative—determines not just individual adherence but the interaction topology between instructions: cooperative in English, competitive in Spanish, flat in French. This is fixable at the prompt level: declare facts, don't issue commands. The fix works, and it spills over to instructions that weren't rewritten.

The mechanism is social register. Models trained on multilingual data have learned that imperative mood carries different obligatory force in different speech communities. When a system prompt stacks imperatives, the resulting interaction topology depends on which language's conventions the model applies. We note that "English" and "Spanish" are proxies here for register-encoding patterns, not language-intrinsic properties—varieties within a language (formal vs. colloquial, regional differences in authority encoding) likely produce different topologies for the same reason.

The inference-time finding has a training-time implication. Constitutional AI principles are written in imperative mood. If register mediates instruction-following at inference time, it plausibly does so during training. Alignment may be language-dependent at the register level. This is a testable prediction. We hope someone tests it.

Total cost of the experimental arc that produced these findings: sixty-nine dollars, verified against billing records. Ninety-six percent of that was one model serving as its own judge.

7 Code and Data Availability

Code, analysis scripts, data files, and reproducibility artifacts are available at `https://github.com/fsgeek/arbiter` (paper snapshot: tag v0.2.0).

Reproduction commands:

```
python scripts/run_cross_linguistic.py --compare
python scripts/run_e_proc.py --compare
python scripts/run_e_topo.py --compare
```

References

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

Yilin Geng, Omri Abend, Eduard Hovy, and Lea Frermann. Measuring pragmatic influence in large language model instructions. *arXiv preprint arXiv:2602.21223*, 2026.

Tony Mason. Arbiter: Detecting interference in LLM agent system prompts. *arXiv preprint arXiv:2603.08993*, 2026.

Itai Mondshine, Tzuf Paz-Argaman, and Reut Tsarfaty. Beyond English: The impact of prompt translation strategies across languages and tasks in multilingual LLMs. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 1331–1354, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-195-7. doi: 10.18653/v1/2025.findings-naacl.73. URL `https://aclanthology.org/2025.findings-naacl.73/`.

Qihan Wang, Shidong Pan, Tal Linzen, and Emily Black. Multilingual prompting for improving llm generation diversity. pages 6378–6400, 2025.

Ziqi Yin, Hao Wang, Kaito Horio, Daisuke Kawahara, and Satoshi Sekine. Should we respect llms? a cross-lingual study on the influence of prompt politeness on llm performance. 2024. URL `https://arxiv.org/abs/2402.14531`.

Lechen Zhang, Yusheng Zhou, Tolga Ergen, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. Cross-lingual prompt steerability: Towards accurate and robust llm behavior across languages. *arXiv preprint arXiv:2512.02841*, 2025.